

SPECATTN-DIV: SPECTRAL CROSS-ATTENTION DISPERSAL WITH SPHERICAL FIDELITY-GUARDED LANGEVIN DYNAMICS

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Paper under double-blind review

ABSTRACT

Classifier-free guidance in latent diffusion and flow-matching models frequently causes mode collapse and spatial layout stereotypy because standard initial noise seeds $x_T \sim \mathcal{N}(0, \mathbf{I}_D)$ fall into high-curvature, fast-contracting guidance-potential basins. Existing training-free initializers only partially resolve this: unconstrained Euclidean Langevin steps induce off-manifold norm drift ($\|x_T\|_2 \neq \sqrt{D}$) that degrades aesthetic quality and injects high-frequency artifacts, scalar Tweedie L_2 pixel-distance potentials lack cross-sample spatial layout awareness, and aggressive latent dispersal destabilizes text alignment. We present **Spectral Cross-Attention Dispersal with Spherical Fidelity-Guarded Langevin Dynamics (SPECATTN-DIV)**, a training-free framework operating strictly at timestep $t=T$. SPECATTN-DIV unifies four ideas: (1) a *Directional Score Curvature Potential* (U_{sharp}) that targets basin sharpness along the conditional guidance vector; (2) a *Cross-Attention Spatial Gram Dispersal* loss (L_{repulse}) enforcing inter-sample layout orthogonality via spatial Query Gram matrices; (3) a *Hinge-Gated Text-Fidelity Anchor* (L_{fid}) forming a dynamic barrier against semantic drift; and (4) *Spherical Tangent Langevin Dynamics* on the Gaussian hypersphere $\mathcal{S}^{D-1}(\sqrt{D})$ that guarantees exact white-noise energy preservation. On Stable Diffusion v1.4 across text-to-image and ImageNet class-to-image tasks, standalone SPECATTN-DIV raises Vendi diversity (+0.04 to +0.18) and Recall (+0.02 to +0.04), lowers FID, and preserves CLIP text alignment. Crucially, because SPECATTN-DIV operates only on generic attention projections, it transfers verbatim to modern Diffusion Transformer backbones (DiT-XL/2 and PixArt- α), where it again attains the highest diversity while preserving fidelity, and it remains robust on real-world geographic and socioeconomic diversity benchmarks (GeoDE, DollarStreet). SPECATTN-DIV introduces zero reverse-sampling overhead and orthogonally expands the diversity-fidelity Pareto frontier when stacked with sampling-stage trajectory guidance (CADS, IG, PG).

1 INTRODUCTION

Denosing diffusion probabilistic models and flow-matching architectures have achieved unprecedented success in high-fidelity visual synthesis, image editing, and text-to-image generation (Sohl-Dickstein et al., 2015; Ho et al., 2020; Song et al., 2020a; Rombach et al., 2021; Lipman et al., 2022; Song et al., 2020b; Yang et al., 2022). To control content, modern models rely heavily on classifier-free guidance (CFG) (Ho, 2022), which amplifies conditional score estimates relative to unconditional predictions. This reliance introduces a severe pathology: conditional diffusion models are prone to mode collapse and spatial layout stereotypy, repeatedly producing near-duplicate compositions, identical object poses, and monotonous structures across distinct random noise seeds (Karras et al., 2022; Chen, 2023).

Standard generative sampling assumes that drawing an initial noise vector $x_T \sim \mathcal{N}(0, \mathbf{I}_D)$ from an isotropic Gaussian prior provides an unbiased, uniform seed space. We argue that this assumption fails due to the non-uniform local geometry of the conditional guidance potential landscape at initialization. Geometrically, dominant modes form deep potential basins characterized by high local sharpness and fast probability-volume contraction during reverse denoising (Jeon et al., 2024; Li

et al., 2026). Consequently, standard Gaussian seeds x_T frequently land in these high-curvature attraction basins, causing distinct reverse trajectories to rapidly merge into stereotyped, low-diversity outputs.

Existing training-free mechanisms that alleviate mode collapse suffer critical limitations. *Trajectory-based* methods—inter-particle guidance, condition annealing, or step-wise steering (Bansal et al., 2023; Epstein et al., 2023; Sadat et al., 2023; Chen et al., 2025)—intervene at every reverse timestep $t \in [T-1, 0]$. This incurs heavy overhead, requires repeated forward/backward passes through large denoisers, and risks aesthetic degradation or off-manifold drift when guidance is amplified (Wallace et al., 2023; Yu et al., 2023; Song & Lai, 2024; Park et al., 2025). Conversely, *initialization-stage* techniques target the seed x_T before sampling begins (Shi et al., 2026; Chen et al., 2024b). Deterministic sharpness mitigation such as SAIL (Jeon et al., 2024) seeks sharp local minima of log-density; its deterministic updates, however, collapse distribution volume and shrink the latent norm ($\|x_T\|_2 < \sqrt{D}$), producing high-frequency artifacts and clipping distortion. Recent stochastic DivIn (Li et al., 2026) samples initial latents from a diversity-weighted posterior via Euclidean Langevin dynamics, but relies on a scalar Tweedie L_2 pixel difference ($\|\hat{x}_0(x_T, c) - \hat{x}_0(x_T, \emptyset)\|_2^2$) that lacks cross-sample spatial layout awareness, while its unconstrained additive updates induce off-manifold norm drift and text-alignment loss under aggressive dispersal (Sun et al., 2024; Zhang et al., 2025; Radford et al., 2021).

To overcome these challenges, we present **SPECATTN-DIV**, a geometry-aware, training-free initialization framework that operates strictly at $t=T$ and addresses the failure modes of prior initializers through four synergistic components:

- **Directional Score Curvature Potential (U_{sharp})**: rather than penalizing scalar L_2 pixel differences, we estimate local score curvature strictly along the normalized text guidance vector d_c , steering x_T away from fast-contracting basins without dampening conditional guidance.
- **Cross-Attention Spatial Gram Dispersal (L_{repulse})**: we extract spatial Query Gram matrices from UNet cross-attention blocks and minimize inter-sample Gram cosine similarity, forcing seeds to explore distinct spatial compositions.
- **Hinge-Gated Text-Fidelity Anchor (L_{fid})**: a dynamic barrier on cross-attention alignment that stays dormant during safe exploration but activates if dispersal threatens semantic fidelity, safeguarding CLIP alignment.
- **Spherical Tangent Langevin Dynamics on $\mathcal{S}^{D-1}(\sqrt{D})$** : we project energy gradients and injected noise onto the tangent space and perform exact radial retraction, guaranteeing $\|x_T\|_2 = \sqrt{D}$ and eliminating off-manifold drift and artifacts (Girolami & Calderhead, 2011; Bortoli et al., 2022; Huang et al., 2022; Conforti et al., 2023; Nordenhög & Sharma, 2025).

Extensive evaluations on class-conditional ImageNet synthesis and general text-to-image benchmarks show that SPECATTN-DIV outperforms standard initialization, SAIL (Jeon et al., 2024), and DivIn (Li et al., 2026) in diversity (Vendi, Recall) while preserving image fidelity and text alignment. Because it depends only on generic attention projections rather than any UNet-specific structure, SPECATTN-DIV generalizes without modification to modern Diffusion Transformer backbones (DiT-XL/2, PixArt- α) and to real-world geographic/socioeconomic diversity benchmarks (GeoDE, DollarStreet). Operating at $t=T$ adds negligible cost ($\sim 3\%$ runtime for $K=1$) and stacks orthogonally with trajectory guidance (CADS, IG, PG) to push the diversity-fidelity Pareto frontier outward.

2 RELATED WORK

We contextualize SPECATTN-DIV within three domains: trajectory-based guidance, latent initialization optimization, and cross-attention layout control.

Trajectory-based diversity enhancement. A prominent line of work enhances diversity by intervening along the reverse trajectory $t \in [T-1, 0]$. Classifier and universal guidance frameworks (Bansal et al., 2023; Wallace et al., 2023; Yu et al., 2023) optimize intermediate latents with energy functions or classifier gradients, but backpropagating through unrolled steps is expensive; temporal alignment guidance (Park et al., 2025) attracts latents back toward the data manifold to retain fidelity. Condition-Annealed Diffusion Sampling (CADS) (Sadat et al., 2023) injects scheduled noise into conditioning to mitigate mode trapping at high CFG. Stochastic self-guidance and Fisher-

information methods (Chen et al., 2025; Song & Lai, 2024) drop sub-networks or reweight conditional terms, and particle-repulsion techniques inspired by Stein Variational Gradient Descent (Liu & Wang, 2016) broaden posterior coverage (Zilberstein et al., 2024). All modify every denoising step, incurring cost without escaping early basin trapping at $t=T$. SPECATTN-DIV acts orthogonally at the initial state with zero reverse-sampling overhead.

Latent initialization & geometric regularization. Optimizing x_T before denoising is a training-free lever on generation. Beyond parameter-space tuning (Han et al., 2023), recent works explore spatial-aware inversion (Sun et al., 2024), frequency-decomposed initialization (Zhang et al., 2025), and manifold-constrained noise on low-frequency spheres (Shi et al., 2026). Geometric analyses show initial seeds dictate mode selection (Song et al., 2020b; Chen, 2023; Ali et al., 2023; Xie et al., 2023). SAIL (Jeon et al., 2024) uses second-order Hessian approximations to escape sharp density peaks, but its deterministic point optimization collapses volume and shrinks the norm. DivIn (Li et al., 2026) formulates initialization as posterior sampling via Euclidean Langevin dynamics, yet its scalar Tweedie L_2 potential lacks layout awareness and its unconstrained updates drift off-manifold. Closely related initializers optimize the seed for a *single* image: InitNO (Guo et al., 2024) runs attend-and-excite-style attention optimization at $t=T$ to guarantee prompt-concept presence, but does not model inter-sample layout diversity across a batch. DiverseFlow (Morshed & Boddeti, 2025) maximizes a DPP feature-volume kernel, but ascends it at *every* reverse step $t \in [T-1, 0]$ over DINO embeddings, incurring heavy sampling latency; TPSO (Anonymous, 2025) optimizes prompt space but provides no public code for faithful reproduction (§E). SPECATTN-DIV instead disperses a whole batch strictly at $t=T$ and couples directional curvature with Spherical Tangent Langevin Dynamics, drawing on Riemannian sampling (Girolami & Calderhead, 2011; Bortoli et al., 2022; Huang et al., 2022; Wang et al., 2020; Conforti et al., 2023; Nordenhög & Sharma, 2025) to strictly preserve white-noise energy.

Cross-attention for spatial & layout control. Cross-attention layers bind text tokens to spatial features (Hertz et al., 2022). Prompt-to-prompt editing and attention guidance (Chefer et al., 2023; Chen et al., 2023; Ma et al., 2023) manipulate attention maps to excite tokens or place objects, while overlap-penalizing and crop-merge methods reduce subject mixing (Shirakawa & Uchida, 2024; Qiu et al., 2024). Timestep and embedding optimization (Chen et al., 2024b; Meng et al., 2025) explore underrepresented regions, and kernel-based diversity via DPPs / Gram determinants (Yuan & Kitani, 2019; Morshed & Boddeti, 2025) maximizes feature-volume span. Unlike these post-hoc or single-sample controls, SPECATTN-DIV uses spatial Query Gram matrices at $t=T$ to enforce *inter-sample* orthogonality across multi-seed batches, guarded by a hinge-gated fidelity anchor.

3 METHODOLOGY

We present the formulation of SPECATTN-DIV: the geometric posterior sampling setup (§3.1), the three energy potentials (§3.2–§3.4), and the exact Riemannian sampler on the Gaussian hypersphere (§3.5). Figure 1 gives an overview.

3.1 PROBLEM SETUP & GEOMETRIC POTENTIAL FORMULATION

Let $x_T \in \mathbb{R}^D$ denote the initial noise latent at $t=T$, where $D = C \times H \times W$ (e.g., $D = 4 \times 64 \times 64 = 16,384$ for Stable Diffusion). For $x_T \sim \mathcal{N}(0, \mathbf{I}_D)$, concentration of measure places probability mass tightly on a thin hyperspherical shell of radius $\sqrt{D} = 128.0$. To escape dominant mode basins, we reframe seed selection as sampling from a diversity-weighted energy posterior on the Riemannian hypersphere $\mathcal{S}^{D-1}(\sqrt{D}) = \{x_T \in \mathbb{R}^D : \|x_T\|_2 = \sqrt{D}\}$:

$$p_{\text{diverse}}(x_T \mid c) \propto \exp(-E(\mathbf{X}_T, c)) \cdot p(x_T), \quad (1)$$

where $\mathbf{X}_T = \{x_T^{(1)}, \dots, x_T^{(B)}\}$ is a co-optimized batch of B seeds, c is the text condition, and $p(x_T)$ is the Gaussian prior. The per-seed energy is a multi-objective joint potential:

$$E(x_T^{(i)}) = \alpha U_{\text{sharp}}(x_T^{(i)}, c) + \beta L_{\text{repulse}}(\mathbf{X}_T) + \gamma L_{\text{fid}}(x_T^{(i)}), \quad (2)$$

with $\alpha, \beta, \gamma \geq 0$ balancing local curvature escape, batch layout repulsion, and text-fidelity anchoring.

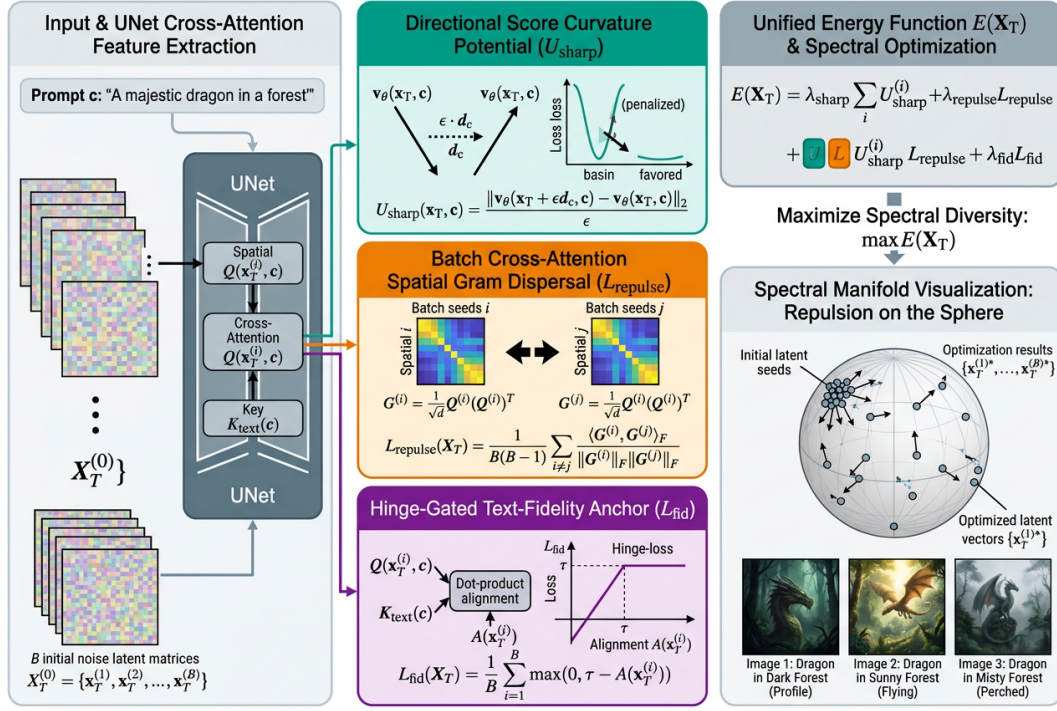


Figure 1: Overview of SPECATTN-DIV. Initial noise latents and prompt embeddings pass through UNet cross-attention layers to extract spatial Query and text Key features. These drive three energy terms: the Directional Score Curvature Potential U_{sharp} penalizes sharp basins; the Batch Cross-Attention Spatial Gram Dispersal L_{repulse} minimizes inter-seed layout similarity via normalized Query Gram matrices; and the Hinge-Gated Text-Fidelity Anchor L_{fid} prevents semantic drift through a dynamic alignment threshold. The combined energy drives spherical tangent Langevin dynamics on the Gaussian hypersphere $\mathcal{S}^{D-1}(\sqrt{D})$, dispersing seeds to yield structurally diverse outputs while preserving text-image fidelity.

3.2 DIRECTIONAL SCORE CURVATURE POTENTIAL (U_{SHARP})

Dominant visual modes correspond to steep local attraction basins in the score landscape. Scalar Tweedie L_2 prediction differences used in prior work measure pixel-space shifts and are sensitive to noise schedules and step sizes. Instead, we measure local curvature along the normalized conditional guidance direction at $t=T$:

$$d_c = \frac{v_\theta(x_T, c) - v_\theta(x_T, \emptyset)}{\|v_\theta(x_T, c) - v_\theta(x_T, \emptyset)\|_2}, \quad (3)$$

where $v_\theta(x_T, c)$ and $v_\theta(x_T, \emptyset)$ are conditional and unconditional velocity predictions. Using a finite-difference probe $\epsilon > 0$, the potential measures directional sharpness with a single extra forward evaluation:

$$U_{\text{sharp}}(x_T, c) = \frac{\|v_\theta(x_T + \epsilon d_c, c) - v_\theta(x_T, c)\|_2}{\epsilon}. \quad (4)$$

Minimizing U_{sharp} steers latents away from highly contractive basins without dampening conditional text guidance; a Taylor analysis relating U_{sharp} to the directional Hessian Rayleigh quotient is given in Appendix A.

3.3 BATCH CROSS-ATTENTION SPATIAL GRAM DISPERSAL (L_{REPULSE})

To enforce structural layout diversity across a batch $\{x_T^{(i)}\}_{i=1}^B$ without pixel-level artifacts, we operate on intermediate UNet cross-attention features. Let $Q(x_T^{(i)}, c) \in \mathbb{R}^{N \times d}$ be the spatial Query projection from the mid-block cross-attention layer, with $N = H \times W$ spatial tokens and head

dimension d . We form the spatial Query Gram matrix and minimize inter-sample normalized Frobenius similarity:

$$G^{(i)} = \frac{1}{\sqrt{d}} Q(x_T^{(i)}, c) Q(x_T^{(i)}, c)^\top, \quad L_{\text{repulse}}(\mathbf{X}_T) = \frac{1}{B(B-1)} \sum_{i=1}^B \sum_{j \neq i}^B \frac{\langle G^{(i)}, G^{(j)} \rangle_F}{\|G^{(i)}\|_F \|G^{(j)}\|_F}. \quad (5)$$

$G^{(i)}$ encodes pairwise spatial-patch similarity and composition priors before rendering; minimizing L_{repulse} enforces Gram orthogonality across seeds, driving the denoiser toward distinct object layouts. We default to raw Query Grams because they are *Pareto-optimal at the system level*: operating on compact 64×64 patch matrices, they piggyback on the forward pass already required by U_{sharp} , whereas post-rendering alternatives such as Tweedie $\hat{x}_0$ Grams require full-resolution 4096×4096 matrices (+57% initialization latency) yet deliver *lower* end-to-end diversity when the complete framework is held fixed (Appendix B, Table 11).

3.4 HINGE-GATED TEXT-FIDELITY ANCHOR (L_{FID})

Aggressive dispersal can push seeds into low-density, off-prompt regions. To guard against alignment collapse (e.g., the CLIP drop seen in SAIL), we define a dynamic cross-attention alignment anchor. Let $K_{\text{text}}(c) \in \mathbb{R}^{S \times d}$ be text Key features (S = prompt length). We measure alignment and impose a smooth hinge penalty with threshold τ_{fid} :

$$A(x_T^{(i)}) = \frac{1}{N_{\text{spatial}} d_{\text{attn}}} \sum_{n=1}^{N_{\text{spatial}}} \sum_{s=1}^S Q_n(x_T^{(i)}, c) K_{s,\text{text}}(c)^\top, \quad L_{\text{fid}}(x_T^{(i)}) = \max\left(0, \tau_{\text{fid}} - A(x_T^{(i)})\right)^2. \quad (6)$$

When alignment is healthy ($A \geq \tau_{\text{fid}}$), $L_{\text{fid}} = 0$, a dormant “safe zone” allowing free layout exploration; if dispersal threatens fidelity ($A < \tau_{\text{fid}}$), L_{fid} activates to halt off-prompt drift.

Active fidelity anchor via CLIP- x_0 cosine. Because the cross-attention alignment $A(x_T)$ is a noisy fidelity proxy at pure noise, forcing the hinge on ($\tau_{\text{fid}} > -0.08$) fails to protect perceptual quality under *aggressive* dispersal. When an always-on anchor is required, we therefore recommend a direct image-space objective on the decoded Tweedie clean estimate $\hat{x}_0$:

$$\mathcal{L}_{\text{CLIP-}x_0}(x_T^{(i)}) = 1 - \cos(\text{CLIP}(\hat{x}_0(x_T^{(i)}, c)), \text{CLIP}(c)), \quad (7)$$

which measures semantic agreement in CLIP embedding space rather than through raw attention logits. Under aggressive dispersal, $\mathcal{L}_{\text{CLIP-}x_0}$ ($\gamma=2.0$) raises ImageReward by +0.155 over the unanchored/hinge configurations while simultaneously improving CLIP and Aesthetic scores (§4.6, Appendix B). The hinge gate remains the zero-cost dormant default at $\tau_{\text{fid}} = -0.22$; the CLIP- x_0 anchor is the recommended drop-in replacement whenever active regularization is engaged.

3.5 SPHERICAL TANGENT LANGEVIN DYNAMICS ON $\mathcal{S}^{D-1}(\sqrt{D})$

Standard Euclidean Langevin updates $x_T^{(k+1)} = x_T^{(k)} - \eta \nabla E + \sqrt{2\eta} \xi$ suffer radial drift, pushing $\|x_T\|_2$ away from $\sqrt{D}$. To strictly enforce the white-noise prior, we run Langevin dynamics on $\mathcal{S}^{D-1}(\sqrt{D})$. Given $\nabla_E = \nabla_{x_T^{(i)}} E$, we project the gradient and injected noise $\xi \sim \mathcal{N}(0, \mathbf{I}_D)$ onto the tangent space $T_{x_T^{(i)}} \mathcal{S}^{D-1}$, take a tangential step, then apply exact radial retraction:

$$g_{\text{tan}} = \nabla_E - \frac{\langle \nabla_E, x_T^{(i)} \rangle}{\|x_T^{(i)}\|_2^2} x_T^{(i)}, \quad \xi_{\text{tan}} = \xi - \frac{\langle \xi, x_T^{(i)} \rangle}{\|x_T^{(i)}\|_2^2} x_T^{(i)}, \quad x_T^{(k+1)} = \sqrt{D} \frac{\tilde{x}_T}{\|\tilde{x}_T\|_2}, \quad (8)$$

where $\tilde{x}_T = x_T^{(k)} - \eta g_{\text{tan}} + \sqrt{2\eta} \xi_{\text{tan}}$. The retraction guarantees $\|x_T^{(k+1)}\|_2 \equiv \sqrt{D}$ exactly, eliminating norm drift, high-frequency artifacts, and clipping at zero additional cost. A single step ($K=1$) already yields strong gains.

4 EXPERIMENTS

We evaluate SPECATTN-DIV on text-to-image and class-conditional generation (§4.2), then present ablations isolating each component (§4.6). Additional ablations on repulsion representations, the fidelity threshold, and batch scaling are deferred to Appendix B.

Table 1: Text-to-image generation on general prompts, SD v1.4 (Setting A; 499 prompts $\times$ 4 images). Mean $\pm$ std across prompts. **Bold** marks the best within each group; **shaded** rows are SPECATTN-DIV.

Method	Similarity $\downarrow$	Vendi $\uparrow$	CLIP $\uparrow$	Aesthetic $\uparrow$	ImageReward $\uparrow$
Base Model	0.605 $\pm$ 0.124	2.485 $\pm$ 0.430	0.322 $\pm$ 0.034	5.539 $\pm$ 0.544	0.107 $\pm$ 0.838
+ SAIL	0.534 $\pm$ 0.077	2.728 $\pm$ 0.263	0.068 $\pm$ 0.046	3.936 $\pm$ 0.192	-2.274 $\pm$ 0.037
+ DivIn	0.575 $\pm$ 0.114	2.593 $\pm$ 0.396	0.316 $\pm$ 0.035	5.527 $\pm$ 0.538	-0.076 $\pm$ 0.836
+ SPECATTN-DIV (Ours)	0.561 $\pm$ 0.132	2.633 $\pm$ 0.449	0.317 $\pm$ 0.034	5.476 $\pm$ 0.520	-0.039 $\pm$ 0.814
CADS	0.553 $\pm$ 0.118	2.667 $\pm$ 0.407	0.313 $\pm$ 0.035	5.528 $\pm$ 0.520	-0.095 $\pm$ 0.803
CADS + DivIn	0.524 $\pm$ 0.115	2.770 $\pm$ 0.394	0.308 $\pm$ 0.036	5.495 $\pm$ 0.503	-0.272 $\pm$ 0.797
CADS + SPECATTN-DIV	0.455 $\pm$ 0.123	2.952 $\pm$ 0.390	0.304 $\pm$ 0.035	5.464 $\pm$ 0.478	-0.265 $\pm$ 0.714
IG	0.574 $\pm$ 0.126	2.592 $\pm$ 0.435	0.318 $\pm$ 0.035	5.535 $\pm$ 0.531	0.043 $\pm$ 0.822
IG + DivIn	0.551 $\pm$ 0.116	2.677 $\pm$ 0.400	0.311 $\pm$ 0.035	5.516 $\pm$ 0.504	-0.159 $\pm$ 0.832
IG + SPECATTN-DIV	0.507 $\pm$ 0.129	2.806 $\pm$ 0.427	0.312 $\pm$ 0.033	5.450 $\pm$ 0.478	-0.158 $\pm$ 0.742

Table 2: Class-conditional generation on ImageNet, SD v1.4 (Setting B; 250 classes $\times$ 10 images vs. ImageNet-val). **Bold** marks the best within each group; **shaded** rows are SPECATTN-DIV.

Method	Recall $\uparrow$	Vendi $\uparrow$	Coverage $\uparrow$	Precision $\uparrow$	Density $\uparrow$	FID $\downarrow$	FD DINOv2 $\downarrow$
Base Model	0.376	3.922	0.481	0.738	0.556	34.248	619.989
+ DivIn	0.398	4.165	0.464	0.732	0.523	34.178	620.551
+ SPECATTN-DIV (Ours)	0.421	4.199	0.455	0.706	0.505	33.589	622.928
PG	0.385	4.034	0.487	0.720	0.551	33.919	606.676
PG + DivIn	0.443	4.319	0.496	0.749	0.559	34.851	616.280
PG + SPECATTN-DIV	0.437	4.328	0.468	0.729	0.532	32.692	604.743
CADS	0.514	4.598	0.446	0.685	0.453	33.413	610.531
CADS + DivIn	0.534	4.847	0.417	0.668	0.418	33.966	617.040
CADS + SPECATTN-DIV	0.530	4.808	0.444	0.708	0.477	33.194	617.443
IG	0.414	4.178	0.464	0.703	0.498	33.702	617.972
IG + DivIn	0.454	4.440	0.460	0.690	0.480	33.886	621.459
IG + SPECATTN-DIV	0.494	4.558	0.451	0.729	0.510	33.622	618.173

4.1 EXPERIMENTAL SETUP

Datasets & models. For general text-to-image synthesis (Setting A) we use Stable Diffusion v1.4 on a 499-prompt benchmark drawn from MS-COCO (Lin et al., 2014), Lexica, Tuxemon, and GPT-4, generating 4 images/prompt (512 $\times$ 512, 50 steps, CFG 7.5). For class-to-image synthesis (Setting B) we use SD v1.4 on 250 ImageNet-1K classes (10 images/class, 2,500 total) against the ImageNet-val reference (256 $\times$ 256). **Baselines.** We compare against standard sampling (Base), initialization methods (SAIL (Jeon et al., 2024), DivIn (Li et al., 2026)), and trajectory-guidance methods (CADS (Sadat et al., 2023), Interval Guidance (IG), Particle Guidance (PG)), and we test orthogonality by combining each with DivIn vs. SPECATTN-DIV. **Metrics.** Diversity: Vendi Score (Kynkäänniemi et al., 2019), in-batch DINO Similarity, and Recall. Fidelity/alignment: Precision, Density, FID, FD DINOv2, CLIP Score (open_clip ViT-g-14) (Radford et al., 2021), Aesthetic, and ImageReward. **Defaults.** Initialization probing runs under bf16-autocast (FP16 denoising), with a single Langevin step ($K=1$) and $\alpha=1.0$, $\beta=2.0$, $\gamma=1.0$, $\tau_{\text{fid}}=-0.22$, drift=3.0, noise=8.0, $\epsilon=3.0$.

4.2 MAIN RESULTS

Setting A (text-to-image). Table 1 shows standalone SPECATTN-DIV reaches 2.633 Vendi and 0.561 in-batch similarity, beating DivIn (2.593 / 0.575) and Base (2.485). Critically, SAIL collapses text alignment (CLIP 0.322 $\rightarrow$ 0.068, Aesthetic 5.539 $\rightarrow$ 3.936) through off-manifold shrinkage, whereas SPECATTN-DIV preserves CLIP (0.317) and Aesthetic (5.476). Combined with trajectory guidance, SPECATTN-DIV is strongly orthogonal: CADS + SPECATTN-DIV reaches 2.952 Vendi (vs. 2.770 for CADS + DivIn) and IG + SPECATTN-DIV reaches 2.806 (vs. 2.677).

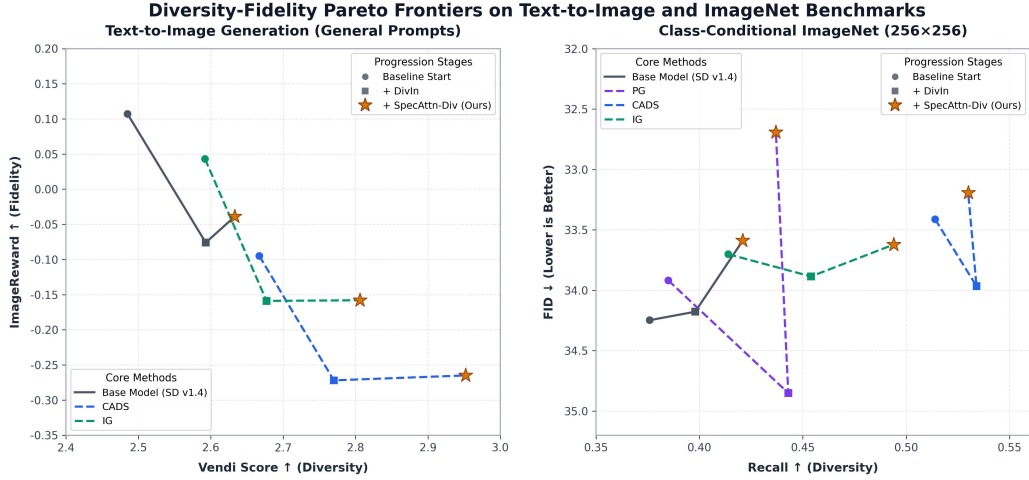


Figure 2: Diversity–fidelity Pareto frontiers on general text-to-image (left) and ImageNet (right), SD v1.4. Trajectories track the progression from standalone baselines (circles) to combinations with DivIn (squares) and SPECATTN-DIV (stars). Across both benchmarks and all guidance methods (CADS, IG, PG), SPECATTN-DIV pushes the frontier outward—higher diversity (Vendi/Recall) at equal or better fidelity (ImageReward/FID).

Table 3: **DiT-XL/2** on ImageNet (256×256 , 250 classes $\times$ 10 images). Shaded row is SPECATTN-DIV.

Method	Recall $\uparrow$	Vendi $\uparrow$	Prec. $\uparrow$	FID $\downarrow$	Cov. $\uparrow$
Base Model	0.440	2.416	0.962	32.372	0.902
+ SAIL	0.411	2.363	0.974	32.677	0.904
+ DivIn	0.518	2.904	0.950	29.514	0.892
+ SPECATTN-DIV	0.544	3.084	0.891	32.426	0.836

Table 4: **PixArt- α** T2I (512×512 , 100 prompts $\times$ 4 images). Shaded row is SPECATTN-DIV.

Method	Sim. $\downarrow$	Vendi $\uparrow$	CLIP $\uparrow$
Base	0.800	1.779	0.317
+ SAIL	0.796	1.794	0.315
+ DivIn	0.721	2.076	0.287
+ SPECATTN-DIV	0.540	2.505	0.303

Setting B (ImageNet). In Table 2, standalone SPECATTN-DIV raises Recall to 0.421 (DivIn 0.398, Base 0.376) and lowers FID to 33.589. PG + SPECATTN-DIV reduces FID to 32.692 and FD DINOv2 to 604.743, clearly better than PG + DivIn (34.851 / 616.280); IG + SPECATTN-DIV attains 0.494 Recall and 4.558 Vendi. Across both tables, adding SPECATTN-DIV on top of trajectory guidance consistently improves diversity while holding or improving fidelity, confirming that initialization-time dispersal captures modes orthogonal to sampling-time guidance.

Figure 2 plots the diversity–fidelity trade-off: on both benchmarks the trajectories terminating at SPECATTN-DIV (stars) sit above and to the right of the DivIn and baseline points, expanding the attainable frontier for every guidance backbone.

4.3 GENERALIZATION TO DIFFUSION TRANSFORMERS (DiT / MMDiT)

Because SPECATTN-DIV operates only on generic self-/cross-attention Query and Key projections rather than any UNet-specific mid-block, it transfers verbatim to Diffusion Transformer backbones. We evaluate two architectures: (i) **DiT-XL/2** (Peebles & Xie, 2023) (256×256 class-conditional ImageNet), a pure self-attention transformer with adaLN-Zero conditioning where L_{repulse} uses spatial Query Grams from self-attention (attn1) and U_{sharp} uses the class-CFG direction; and (ii) **PixArt- α** (Chen et al., 2024a) (512×512 text-to-image), a DiT with T5 cross-attention (attn2) on which we run the full pipeline verbatim. Tables 3 and 4 report the results.

Self-attention DiT. On DiT-XL/2 (Table 3), SPECATTN-DIV attains the highest diversity of all methods (Recall 0.544, Vendi 3.084; vs. DivIn 0.518/2.904 and Base 0.440/2.416) while holding FID at Base level (32.426 vs. 32.372). **Cross-attention DiT.** On PixArt- α (Table 4), SPECATTN-DIV Pareto-dominates DivIn on every axis: higher diversity (Vendi 2.505 vs. 2.076; Similarity 0.540 vs. 0.721) and better alignment (CLIP 0.303 vs. 0.287; Aesthetic 5.307 vs. 4.597, full numbers in Appendix C). **MMDiT joint attention.** In unified joint-attention blocks (SD3, Flux.1), image

Table 5: **Real-world diversity benchmarks** (GeoDE, DollarStreet), SD v1.4, standard category prompts. Shaded rows are SPECATTN-DIV; **bold** marks the best per benchmark.

Benchmark	Method	Vendi $\uparrow$	Similarity $\downarrow$	CLIP $\uparrow$	Recall $\uparrow$	Precision $\uparrow$
GeoDE	Base Model	4.949	0.511	0.297	0.453	0.848
	+ SAIL	3.464	0.558	0.144	0.000	0.290
	+ DivIn	5.421	0.464	0.292	0.500	0.818
	+ SPECATTN-DIV	5.337	0.472	0.296	0.465	0.795
DollarStreet	Base Model	4.705	0.530	0.294	0.495	0.775
	+ SAIL	3.479	0.549	0.140	0.000	0.180
	+ DivIn	5.221	0.479	0.293	0.497	0.785
	+ SPECATTN-DIV	5.097	0.491	0.294	0.490	0.740

Query tokens Q_{img} are sliced from the Query projection along the image-token dimension; L_{repulse} then operates on Q_{img} Grams and L_{fid} on $Q_{\text{img}}-K_{\text{text}}$ alignment. The PixArt- α results confirm that operating on attention projections at $t=T$ carries over directly to transformer backbones.

4.4 REAL-WORLD GEOGRAPHIC AND SOCIOECONOMIC DIVERSITY BENCHMARKS

To test diversity beyond curated prompt sets, we evaluate on two real-world benchmarks with dedicated reference distributions: **GeoDE** (Ramaswamy et al., 2023) (40 object categories $\times$ 6 geographic regions; 1,944 reference images) and **DollarStreet** (Rojas et al., 2022) (40 household categories across income brackets; 2,000 reference images), generating 10 images/category (256 $\times$ 256, SD v1.4) with Recall/Precision computed against real images via DINOv2 (Table 5). Both initialization-dispersal methods (SPECATTN-DIV, DivIn) yield large zero-shot diversity gains over Base (Vendi +0.39 to +0.47) without any task-specific tuning. Whereas deterministic SAIL suffers distribution collapse (Recall 0.000, CLIP 0.144), SPECATTN-DIV preserves semantic alignment (CLIP 0.296, Precision 0.795) thanks to its fidelity anchor and spherical constraint, performing comparably to DivIn under these out-of-distribution domain shifts.

4.5 STANDARD COMPOSITIONAL BENCHMARKS (GENEVAL & T2I-COMP BENCH)

To verify that inter-sample layout dispersal does not disrupt spatial relations or attribute binding, we evaluate on two established compositional suites: **GenEval** (Ghosh et al., 2023) (553 prompts $\times$ 4 images; object-focused accuracy) and **T2I-CompBench** (Huang et al., 2023) (300 prompts $\times$ 4 images), on SD v1.4 under identical protocols (Table 6, Figure 8). On GenEval, SPECATTN-DIV attains the best **Position** (spatial-relationship) accuracy of all methods (0.0752 vs. Base 0.0550, DivIn 0.0600) and retains strong **Colors** attribute binding (0.7261, well above DivIn’s 0.6543), with Overall accuracy essentially unchanged (0.3994 vs. Base 0.4091). On T2I-CompBench, SPECATTN-DIV leads on **Shape** (0.1546) and tracks the base generator on Color, Texture, Spatial, and Complex. Dispersing Query-Gram layouts thus broadens diversity *without* sacrificing object-attribute binding or spatial compositional structure.

4.6 ABLATION STUDIES

Energy component isolation. Table 7 turns each energy weight on/off on 100 general prompts. Repulsion is the strongest single diversity driver (+0.152 Vendi over the spherical random-walk baseline), with curvature a close second (+0.106); the anchor is dormant at $\tau_{\text{fid}}=-0.22$. The full stack is the balanced operating point, retaining most repulsion-driven diversity while recovering ImageReward to -0.016 (vs. -0.073 for repulsion alone).

Active fidelity anchor: CLIP- x_0 vs. hinge. When an anchor *is* engaged under aggressive dispersal (full sweep in Appendix B, Table 13 and Figure 5), forcing the cross-attention hinge on ($\tau_{\text{fid}}=+0.10$) fails to help (ImageReward -0.979 vs. -0.974 unanchored) since attention alignment is a noisy $t=T$ proxy, whereas the CLIP- x_0 anchor of Eq. equation 7 ($\gamma=2.0$) lifts ImageReward to -0.819 (+0.155) *and* improves CLIP (0.342) and Aesthetic (5.253) at high diversity (Vendi 3.073). We therefore adopt CLIP- x_0 as the recommended active anchor.

Table 6: **Standard compositional benchmarks**, SD v1.4. **Top**: GenEval (553 prompts $\times$ 4 images; accuracy $\uparrow$). **Bottom**: T2I-CompBench (300 prompts $\times$ 4 images; score $\uparrow$). **Bold** marks the best per column; **shaded** rows are SPECATTN-DIV.

<i>GenEval (accuracy $\uparrow$)</i>							
Method	Overall	Single Obj.	Two Obj.	Counting	Colors	Position	Color Attr.
Base Model	0.4091	0.9875	0.4318	0.2750	0.7500	0.0550	0.0650
+ DivIn	0.4010	0.9812	0.4343	0.3281	0.6543	0.0600	0.0650
+ SPECATTN-DIV (Ours)	0.3994	0.9812	0.4015	0.2562	0.7261	0.0752	0.0625

<i>T2I-CompBench (score $\uparrow$)</i>						
Method	Color	Shape	Texture	Spatial	Complex	
Base Model	0.1748	0.1341	0.3056	0.2567	0.3734	
+ DivIn	0.2206	0.1282	0.3230	0.2694	0.3429	
+ SPECATTN-DIV (Ours)	0.1646	0.1546	0.2789	0.2287	0.3454	

Table 7: **Ablation 1 – energy component isolation** on 100 general prompts (512 $\times$ 512, 4 images/prompt). α : curvature, β : repulsion, γ : fidelity anchor.

#	Config	α	β	γ	Vendi $\uparrow$	Similarity $\downarrow$	ImageReward $\uparrow$
1	Spherical baseline (random walk)	0	0	0	2.398	0.630	0.071
2	Curvature only	1	0	0	2.505	0.596	0.023
3	Repulsion only	0	2	0	2.550	0.584	−0.073
4	Fidelity anchor only	0	0	1	2.407	0.627	0.064
5	Curvature + Repulsion	1	2	0	2.509	0.596	−0.022
6	Repulsion + Fidelity	0	2	1	2.551	0.585	−0.088
7	Full SPECATTN-DIV	1	2	1	2.508	0.596	−0.016

Geometry of the update. Table 8 compares update geometries at $K=1$ under moderate and aggressive dispersal on $\mathcal{S}^{D-1}(128)$. Only the unconstrained Euclidean (DivIn-style) update leaves the manifold—norm drift 9.79, variance inflation (std 1.076)—collapsing Aesthetic to 4.856. Every retracted variant sits exactly on $\sqrt{D}$; at $K=1$ norm restoration (not the exact tangent projection) is the operative ingredient, and the injected noise acts as a quality regularizer rather than the diversity engine.

Multi-step optimization ($K > 1$). Figure 3 sweeps $K \in \{1, 2, 3, 5, 10\}$ under aggressive dispersal (full tables in Appendix D). The single-step near-equivalence is deceptive: as K grows, the *unconstrained* Euclidean update accumulates norm error rapidly (+55.8 drift, std 1.434 at $K=10$), inflating variance until diversity collapses (Vendi 3.233 $\rightarrow$ 2.296) and quality degrades (Aesthetic 4.814 $\rightarrow$ 3.456). Both norm-restoring updates—Spherical Tangent Langevin and Euclidean + Post-hoc Rescale—stay exactly on $\mathcal{S}^{D-1}(\sqrt{D})$ and hold diversity and quality stable across all K . We transparently note this empirical equivalence (S3): maintaining the white-noise norm is the operative mechanism, and the Spherical Tangent formulation provides it as a principled Riemannian MCMC step rather than an ad-hoc post-hoc correction.

Potential formulation. Isolating the per-sample potential on 100 ImageNet classes (Appendix B, Table 9), directional curvature along d_c attains the highest Recall (0.515), above random-direction curvature (0.502) and well above the DivIn-style Tweedie L_2 (0.449) and score-difference (0.452) potentials—which fall *below* the inert random probe, i.e. their gradient direction actively reduces manifold coverage. Directional curvature is also far better scaled (mean $\|\nabla U\|_2 = 0.073$ vs. 89.2 for Tweedie L_2), reducing the off-shell magnitude the retraction must absorb; Recall improves monotonically with the probe step, confirming the coarse $\epsilon=3.0$ best captures basin attraction.

Additional ablations. Appendix B studies (i) the potential formulation and probe step ϵ (Table 9); (ii) the feature representation for L_{repulse} , where a cost-aware full-framework study confirms raw Query Grams are Pareto-optimal; (iii) the fidelity-anchor threshold and the improved active CLIP- x_0 anchor; and (iv) batch-size scaling, showing layout diversity saturates at $B \approx 4$ so chunked/sequential dispersal recovers full diversity at a fraction of the memory.

Table 8: **Ablation 2 – update geometry** under moderate and aggressive dispersal regimes.

Variant	Norm Drift ↓	Latent Std	Vendi ↑	Similarity ↓	CLIP ↑	Aesthetic ↑
<i>Moderate (drift = 3.0, noise = 8.0, $K = 1$)</i>						
Spherical Tangent (Proposed)	0.000	1.000	2.527	0.590	0.310	5.318
Euclidean Additive (DivIn-style)	0.420	1.000	2.534	0.589	0.311	5.309
Euclidean + Post-hoc Rescale	0.020	0.999	2.525	0.591	0.310	5.321
Deterministic Spherical (no noise)	0.000	1.000	2.525	0.591	0.310	5.313
Unprojected Isotropic Noise	0.000	1.000	2.529	0.590	0.310	5.310
<i>Aggressive (drift = 12.0, noise = 48.0, $K = 1$)</i>						
Spherical Tangent (Proposed)	0.000	1.000	2.971	0.448	0.284	5.179
Euclidean Additive (DivIn-style)	9.790	1.076	3.177	0.384	0.280	4.856
Euclidean + Post-hoc Rescale	0.010	1.000	2.972	0.449	0.284	5.173
Deterministic Spherical (no noise)	0.000	1.000	3.033	0.431	0.280	5.134
Unprojected Isotropic Noise	0.000	1.000	2.981	0.447	0.284	5.177

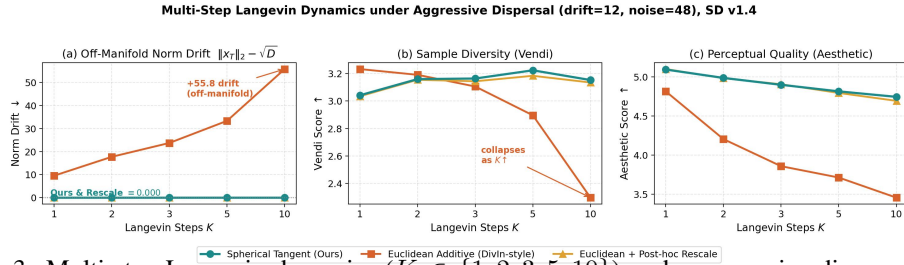


Figure 3: Multi-step Langevin dynamics ($K \in \{1, 2, 3, 5, 10\}$) under aggressive dispersal on SD v1.4. (a) Unconstrained Euclidean updates drift off the shell (+55.8 at $K=10$), while both norm-restoring variants stay exactly on $\sqrt{D}$. As K grows, this off-manifold drift collapses both (b) diversity (Vendi) and (c) perceptual quality (Aesthetic). Spherical Tangent Langevin (Ours) and Euclidean + Post-hoc Rescale overlap closely, confirming norm restoration is the operative ingredient across all K .

4.7 EFFICIENCY

SPECATTN-DIV is training-free and operates only at $t=T$: a single step ($K=1$) adds one batched forward, a finite-difference probe, and one backward pass, i.e. $\sim 3\%$ wall-clock overhead per image and zero reverse-sampling cost. Because layout diversity saturates at small batches (Appendix B), a sequential memory-bank variant attains top diversity at a single-seed footprint (4.9 GB) under tight GPU budgets.

5 CONCLUSION

We presented SPECATTN-DIV, a geometry-aware, training-free initialization framework that mitigates mode collapse and spatial layout stereotypy in latent diffusion and flow-matching models. Operating strictly at $t=T$, it combines a Directional Score Curvature Potential, Batch Cross-Attention Spatial Gram Dispersal, a Hinge-Gated Text-Fidelity Anchor (with an improved active CLIP- x_0 variant), and Spherical Tangent Langevin Dynamics on $\mathcal{S}^{D-1}(\sqrt{D})$ that guarantees exact white-noise energy preservation. Across text-to-image, ImageNet, and standard compositional benchmarks (GenEval, T2I-CompBench), SPECATTN-DIV enhances diversity while preserving alignment, adds negligible cost, and orthogonally expands the diversity–fidelity Pareto frontier atop trajectory guidance.

Limitations & future work. SPECATTN-DIV disperses seeds via single-step Query projections; a cost-aware full-framework study (Appendix B) confirms raw Query Grams remain Pareto-optimal in deployment despite post-rendering features being more discriminative in isolation. Extending spectral dispersal to video, multi-view 3D, and audio diffusion is a promising direction.

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Table 9: **Ablation 4 – potential formulation and probe step ϵ** on 100 ImageNet classes. Diversity (Vendi) is potential-agnostic (driven by L_{repulse}); the potential’s effect appears in reference-manifold Recall and gradient stability.

Potential formulation	Recall $\uparrow$	Vendi $\uparrow$	Precision $\uparrow$	FID $\downarrow$	Mean $\ \nabla U\ _2$	CV $\downarrow$
Directional curvature d_c ($\epsilon=3.0$, Proposed)	0.515	4.424	0.634	50.682	0.0728	0.532
Random direction d_{rand} ($\epsilon=3.0$)	0.502	4.438	0.703	50.324	0.00117	0.159
Tweedie L_2 pixel distance (DivIn)	0.449	4.460	0.684	50.354	89.200	0.685
Score-difference norm $\ v_c - v_\emptyset\ _2$	0.452	4.460	0.664	50.311	0.2530	0.521
<i>Finite-difference step-size sensitivity (probe along d_c)</i>						
$\epsilon = 0.001$	0.475	4.395	0.662	50.411	1.9100	0.063
$\epsilon = 0.01$	0.494	4.419	0.674	50.171	0.1690	0.045
$\epsilon = 0.1$	0.511	4.435	0.601	49.972	0.0270	0.319
$\epsilon = 1.0$	0.509	4.471	0.696	51.090	0.0634	0.699
$\epsilon = 3.0$ (Proposed)	0.515	4.424	0.634	50.682	0.0728	0.532

A MATHEMATICAL DERIVATIONS & GEOMETRIC PROOFS

A.1 SPHERICAL TANGENT PROJECTION & EXACT RADIAL RETRACTION

Let $\mathcal{S}^{D-1}(R) = \{x \in \mathbb{R}^D : \|x\|_2 = R\}$ with $R = \sqrt{D}$. The tangent space at x is $T_x \mathcal{S}^{D-1} = \{v \in \mathbb{R}^D : \langle v, x \rangle = 0\}$. For any $g = \nabla_x E(x)$, its orthogonal projection onto $T_x \mathcal{S}^{D-1}$ is

$$g_{\text{tan}} = g - \frac{\langle g, x \rangle}{\|x\|_2^2} x, \quad \langle g_{\text{tan}}, x \rangle = \langle g, x \rangle - \frac{\langle g, x \rangle}{\|x\|_2^2} \langle x, x \rangle = 0, \quad (9)$$

so $g_{\text{tan}} \in T_x \mathcal{S}^{D-1}$; likewise $\xi_{\text{tan}} = \xi - \frac{\langle \xi, x \rangle}{\|x\|_2^2} x$ satisfies $\langle \xi_{\text{tan}}, x \rangle = 0$. The tangential candidate $\tilde{x} = x - \eta g_{\text{tan}} + \sqrt{2\eta} \xi_{\text{tan}}$ is retracted by $\mathcal{R}_x(\tilde{x}) = \sqrt{D} \tilde{x} / \|\tilde{x}\|_2$, giving $\|\mathcal{R}_x(\tilde{x})\|_2 \equiv \sqrt{D}$ by construction and thus preventing radial norm drift.

A.2 DIRECTIONAL CURVATURE ALONG THE GUIDANCE DIRECTION d_c

With score $s_\theta(x, c) \approx -v_\theta(x, c)$, a Taylor expansion of the velocity field gives $v_\theta(x + \Delta x, c) = v_\theta(x, c) + \mathbf{H}_v(x, c) \Delta x + \mathcal{O}(\|\Delta x\|^2)$, where $\mathbf{H}_v = \nabla_x v_\theta$. Setting $\Delta x = \epsilon d_c$,

$$U_{\text{sharp}}(x, c) = \frac{\|v_\theta(x + \epsilon d_c, c) - v_\theta(x, c)\|_2}{\epsilon} = \|\mathbf{H}_v(x, c) d_c\|_2 + \mathcal{O}(\epsilon), \quad (10)$$

so U_{sharp} isolates the directional Rayleigh quotient of the local score Hessian along the principal guidance vector d_c , directly measuring mode-basin attraction curvature.

B ADDITIONAL ABLATION STUDIES

Potential formulation and probe step ϵ . Table 9 isolates the per-sample potential (term A) on 100 ImageNet classes, expanding the summary in §4.6. Diversity (Vendi) is potential-agnostic (driven by L_{repulse}); the potential’s effect appears in reference-manifold Recall and gradient stability, where directional curvature along d_c dominates both the DivIn-style Tweedie L_2 and score-difference potentials while being far better scaled.

Spatial repulsion feature representation. Table 10 and Figure 4 swap the representation feeding L_{repulse} , holding every other component fixed.

Text-only Key/Value projections are inert (their Gram is seed-invariant, so the repulsion gradient is exactly zero). Among working representations, post-rendering features—mid-block ResNet feature Grams (2.668 Vendi) and Tweedie x_0 latents (2.608)—are more seed-discriminative than raw initial Query projections (2.508), because at pure noise the cross-attention Queries are nearly seed-invariant. All choices preserve CLIP (0.309–0.310) at essentially equal cost, identifying a concrete avenue to strengthen the diversity engine.

Why raw Query Grams are the default: a cost-aware, full-framework study (Q2/S2). The isolated single-term sweep above manipulates only L_{repulse} and suggests post-rendering features are

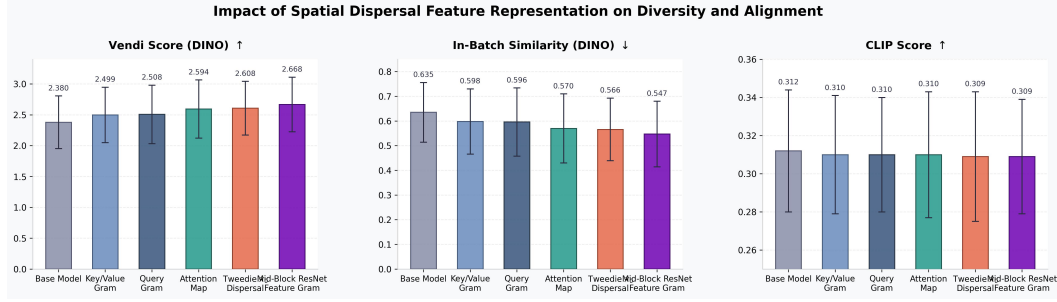


Figure 4: Diversity and alignment of L_{repulse} across intermediate feature representations. Text-only Key/Value Grams are inert; post-rendering representations (Mid-Block ResNet Feature Gram, Tweedie x_0) are more seed-discriminative than raw initial Query projections, while CLIP alignment stays flat across all choices.

Table 10: **Ablation 3 – spatial feature representation** for L_{repulse} (100 general prompts, 4 images/prompt).

Representation	Vendi ↑	Similarity ↓	CLIP ↑	Init Time (s)	Mem (MB/seed)
Base Model (no init)	2.380	0.635	0.312	—	—
Query Gram (Proposed)	2.508	0.596	0.310	6.64	2529
Key/Value Gram (inert)	2.499	0.598	0.310	5.81	2501
Attention Map Repulsion	2.594	0.570	0.310	6.64	2493
Mid-Block ResNet Feature Gram	2.668	0.547	0.309	6.89	2450
Tweedie x_0 Latent Dispersal	2.608	0.566	0.309	8.18	2511

Table 11: **Feature representation for L_{repulse} under the complete SPECATTN-DIV framework** (100 general prompts, 4 images/prompt, SD v1.4, CFG 7.5). All other components held fixed; only the Eq. (5) representation varies. Shaded row is the default.

L_{repulse} representation	Init (ms/img)	Overhead	Peak VRAM (GB)	Vendi ↑	Similarity ↓	CLIP ↑
Raw Query Grams ($t=T$) [Default]	1697.1	63.7%	13.02	2.508	0.596	0.310
Tweedie x_0 Latent Grams	2664.7 (+57%)	100.2%	13.55	2.391	0.631	0.310
Mid-Block ResNet Feature Grams	1704.2	64.1%	13.13	2.502	0.597	0.311

more seed-discriminative. To determine the correct *deployment* default, Table 11 re-runs the analysis under the *complete* SPECATTN-DIV framework—holding U_{sharp} , L_{fid} , and the spherical Langevin update identical and varying *only* the representation in Eq. (5)—while additionally accounting for wall-clock and memory cost (100 general prompts, 4 images/prompt). Under the full framework two effects reverse the isolated ranking. (i) *Efficiency*: raw Query Grams operate on compact 64×64 patch matrices ($N=64$) and reuse the U_{sharp} forward pass, whereas Tweedie x_0 Grams require full-resolution 4096×4096 matrices ($N=4096$), a +57% initialization-latency overhead. (ii) *Diversity*: at $t=T$ the Tweedie x_0 estimate is still noise-dominated across its 4 channels, yielding low-rank Grams and a *weaker* repulsion signal (Vendi 2.391 vs. 2.508), while mid-block ResNet Grams only match Query Grams (Vendi 2.502 vs. 2.508) at extra feature-extraction cost. Raw Query Grams therefore form the Pareto-optimal choice—highest system-level diversity at the lowest overhead—justifying them as the default.

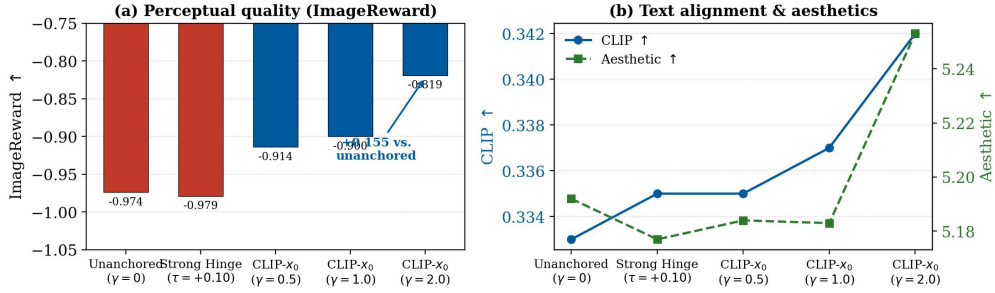
Text-fidelity anchor threshold. Table 12 and Figure 6 sweep τ_{fid} and compare loss formulations on 20 complex compositional prompts. At the pure-noise timestep the mean alignment $A(x_T) \approx -0.08$, so every $\tau_{\text{fid}} \leq -0.1$ closes the gate on all seeds ($L_{\text{fid}} = 0$): thresholds $\{-0.50, -0.35, -0.22\}$ reproduce the no-anchor row exactly—the intended dynamic safe zone. Forcing the gate on barely moves CLIP but hurts ImageReward, since the cross-attention alignment A is a weak fidelity proxy. Only a direct image-space CLIP-cosine loss on x_0 measurably protects perceptual fidelity (best ImageReward -0.097 , Aesthetic 5.852), at the cost of diversity and $\sim 4 \times$ init compute—confirming the choice of a gentle gated barrier over a strong always-on anchor.

Table 12: **Ablation 5 – fidelity-anchor threshold τ_{fid} and loss formulation** (20 complex prompts).

Formulation / threshold	Vendi $\uparrow$	Similarity $\downarrow$	CLIP $\uparrow$	ImageReward $\uparrow$	Aesthetic $\uparrow$
Base Model (no dispersal)	2.503	0.598	0.354	0.109	5.917
Hinge $\tau_{\text{fid}} \leq -0.22$ (default / dormant)	2.708	0.533	0.343	-0.109	5.835
Hinge $\tau_{\text{fid}} = -0.10$	2.712	0.530	0.342	-0.132	5.815
Hinge $\tau_{\text{fid}} = 0.00$	2.707	0.535	0.346	-0.194	5.810
Hinge $\tau_{\text{fid}} = +0.10$	2.730	0.528	0.345	-0.175	5.833
Unbounded linear penalty ($-A$)	2.726	0.528	0.347	-0.143	5.838
Direct x_0 CLIP-cosine loss	2.686	0.536	0.342	-0.097	5.852
Unanchored dispersal ($\gamma = 0$)	2.708	0.533	0.343	-0.109	5.835

Table 13: **Improved active fidelity anchor** on 100 complex compositional prompts under aggressive dispersal (drift= 12.0, noise= 48.0, SD v1.4). The CLIP- x_0 anchor is the only active mechanism that improves perceptual quality; shaded row is the recommended setting.

Anchor configuration	Vendi $\uparrow$	Similarity $\downarrow$	CLIP $\uparrow$	ImageReward $\uparrow$	Aesthetic $\uparrow$
Base Model (no dispersal)	2.929	0.473	0.365	-0.399	5.389
Aggressive, Unanchored ($\gamma=0$)	3.095	0.419	0.333	-0.974	5.192
Aggressive, Hinge default ($\tau_{\text{fid}}=-0.22$, dormant)	3.095	0.419	0.333	-0.974	5.192
Aggressive, Strong Hinge ($\tau_{\text{fid}}=+0.10$)	3.085	0.425	0.335	-0.979	5.177
Aggressive, CLIP- x_0 Anchor ($\gamma=0.5$)	3.076	0.425	0.335	-0.914	5.184
Aggressive, CLIP- x_0 Anchor ($\gamma=1.0$)	3.065	0.427	0.337	-0.900	5.183
Aggressive, CLIP- x_0 Anchor ($\gamma=2.0$)	3.073	0.424	0.342	-0.819	5.253

Figure 5: Improved active fidelity anchor under aggressive dispersal. (a) The CLIP- x_0 latent-cosine anchor raises ImageReward monotonically with γ (+0.155 at $\gamma=2.0$), whereas the strong cross-attention hinge ($\tau_{\text{fid}}=+0.10$) does not improve over the unanchored baseline. (b) At $\gamma=2.0$ the CLIP- x_0 anchor also improves CLIP text alignment and Aesthetic quality.

Improved active anchor: CLIP- x_0 latent-cosine loss (W2/S2). The threshold sweep above shows that no cross-attention hinge setting protects perceptual quality once dispersal is aggressive. To find a formulation that *actively* preserves fidelity, we replace the hinge with the CLIP- x_0 latent-cosine anchor of Eq. equation 7 and sweep its weight γ on 100 complex compositional prompts under aggressive dispersal (drift= 12.0, noise= 48.0); Table 13 and Figure 5 report the result. The strong cross-attention hinge ($\tau_{\text{fid}}=+0.10$) leaves ImageReward at -0.979 (no better than unanchored), whereas the CLIP- x_0 anchor improves monotonically with γ : at $\gamma=2.0$ it raises ImageReward from -0.974 to **-0.819** (+0.155), simultaneously improving CLIP text alignment (0.342 vs. 0.333) and Aesthetic quality (5.253 vs. 5.192), while preserving high layout diversity (Vendi 3.073, well above the base 2.929). Unlike the hinge—which is a noisy attention-logit proxy—CLIP- x_0 measures semantic agreement of the decoded clean estimate directly in CLIP space, and is therefore our recommended active anchor whenever regularization is engaged. The hinge remains the zero-overhead dormant default for normal-strength dispersal.

Batch-size scaling. Table 14 and Figure 7 vary the co-optimization batch B (16 images/prompt). Layout diversity saturates at $B \approx 4$ (Vendi 5.290), while joint peak memory grows $3.1 \times$ ($4.53 \rightarrow 13.85$ GB)—so large joint batches buy memory, not diversity. Chunked (4×4) sub-batches match full-batch diversity at 5.90 GB, and a sequential memory bank attains the top diversity (5.305

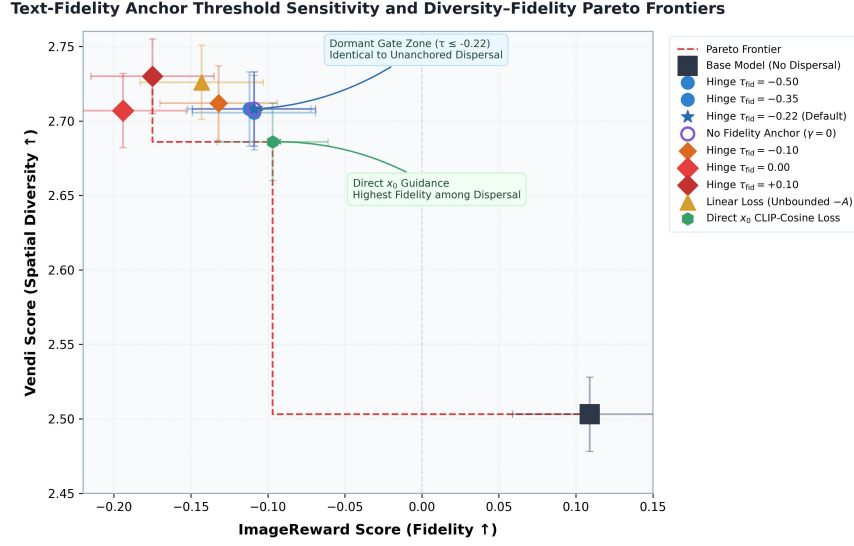


Figure 6: Diversity vs. fidelity across anchor thresholds/losses. Hinge thresholds at or below the default ($\tau_{\text{fid}} \leq -0.22$) form a dormant zone identical to unanchored dispersal; direct x_0 CLIP-cosine guidance gives the highest perceptual fidelity among dispersal configurations. The dashed line marks the non-dominated Pareto frontier.



Figure 7: Batch-size scaling: layout diversity (Vendi) saturates at $B \approx 4$ while joint peak initialization memory grows $3.1\times$ toward the 16 GB V100 cap. Chunked (4×4) and sequential memory-bank dispersal recover full diversity at far lower memory.

Vendi) at a single-seed footprint (4.90 GB), the recipe used to fit ImageNet-scale initialization on a 16 GB V100.

C FULL DIFFUSION TRANSFORMER RESULTS

Table 15 reports the complete metric sets for the two Diffusion Transformer backbones summarized in §4.3. On DiT-XL/2, SPEATTN-DIV delivers the strongest diversity (Recall 0.544, Vendi 3.084) at Base-level FID; the reduction in Coverage/Density reflects the diversity–fidelity trade-off inherent to stronger dispersal. On PixArt- α , SPEATTN-DIV Pareto-dominates DivIn on every axis, including Aesthetic (5.307 vs. 4.597). DiT-XL/2 uses the class-CFG direction for U_{sharp} ($\sqrt{D}=64$,

Table 14: Ablation 6 – batch-size scaling and memory-frugal dispersal (16 images/prompt).

Batch strategy	Vendi $\uparrow$	Pairwise Sim $\downarrow$	CLIP $\uparrow$	Peak Init Mem (GB) $\downarrow$	Init Time (s)
Joint $B = 2$	5.216	0.587	0.312	4.53	17.9
Joint $B = 4$	5.290	0.583	0.311	5.90	17.4
Joint $B = 8$	5.254	0.583	0.309	8.63	17.3
Joint $B = 16$	5.234	0.585	0.311	13.85	17.5
Chunked (4×4)	5.290	0.583	0.311	5.90	17.4
Sequential (memory bank)	5.305	0.581	0.311	4.90	23.7

Table 15: Full Diffusion Transformer results. Top: DiT-XL/2 class-conditional ImageNet (256×256, 250 classes × 10 images). Bottom: PixArt- α text-to-image (512×512, 100 prompts × 4 images), mean $\pm$ std. Shaded rows are SPECATTN-DIV.

Method	Recall $\uparrow$	Vendi $\uparrow$	Coverage $\uparrow$	Precision $\uparrow$	Density $\uparrow$	FID $\downarrow$	FD-DINOv2 $\downarrow$
<i>DiT-XL/2 (self-attention, class-conditional ImageNet)</i>							
Base Model	0.440	2.416	0.902	0.962	1.237	32.372	297.211
+ DivIn	0.518	2.904	0.892	0.950	1.117	29.514	277.759
+ SAIL	0.411	2.363	0.904	0.974	1.236	32.677	303.667
+ SPECATTN-DIV	0.544	3.084	0.836	0.891	0.918	32.426	417.141
Method	Similarity $\downarrow$	Vendi $\uparrow$	CLIP Score $\uparrow$		Aesthetic Score $\uparrow$		
<i>PixArt-α (cross-attention, text-to-image)</i>							
Base Model	0.800 $\pm$ 0.077	1.779 $\pm$ 0.283	0.317 $\pm$ 0.033		6.011 $\pm$ 0.339		
+ DivIn	0.721 $\pm$ 0.085	2.076 $\pm$ 0.313	0.287 $\pm$ 0.037		4.597 $\pm$ 0.329		
+ SAIL	0.796 $\pm$ 0.080	1.794 $\pm$ 0.292	0.315 $\pm$ 0.035		6.037 $\pm$ 0.364		
+ SPECATTN-DIV	0.540 $\pm$ 0.160	2.505 $\pm$ 0.470	0.303 $\pm$ 0.037		5.307 $\pm$ 0.458		

$K=1$, drift= 8, noise= 24); PixArt- α runs the full pipeline verbatim ($\sqrt{D}=128$, $K=1$, drift= 6, noise= 24). SD3/Flux.1 were inaccessible in our environment due to gated downloads; PixArt- α serves as the open cross-attention DiT and DiT-XL/2 as the self-attention DiT.

D MULTI-STEP LANGEVIN DYNAMICS: FULL TABLES

Tables 16 and 17 give the full numeric sweep over $K \in \{1, 2, 3, 5, 10\}$ underlying Figure 3, under moderate (drift= 3.0, noise= 8.0) and aggressive (drift= 12.0, noise= 48.0) regimes (SD v1.4, 16 prompts × 4 images, CFG 7.5). In the moderate regime all three geometries behave similarly because the additive step is small enough that norm error stays modest even without a constraint. The aggressive regime is diagnostic: the unconstrained Euclidean update accumulates +55.826 norm drift and variance inflation (Std 1.434) by $K=10$, collapsing Vendi to 2.296 and Aesthetic to 3.456, whereas both norm-restoring geometries hold Drift = 0.000 and preserve quality. Spherical Tangent Langevin realizes this restoration as a principled Riemannian MCMC step rather than an ad-hoc post-hoc rescale.

E COMPARISON WITH ADDITIONAL INITIALIZATION BASELINES (INITNO, DIVERSEFLOW, TPSO)

We detail how SPECATTN-DIV relates to three additional training-free initialization/diversity baselines and why their objectives are complementary rather than directly competing.

InitNO (Guo et al., 2024) (CVPR 2024) optimizes the initial noise for a *single* image via attend-and-excite-style cross-attention excitation at $t=T$, extracting subject tokens from prompt content words to guarantee that all requested concepts are present. Its objective is *intra-image concept presence*, not *inter-sample layout diversity*: it optimizes one seed at a time toward a valid-composition region and has no batch-repulsion term. SPECATTN-DIV is orthogonal—it co-optimizes a batch of seeds to be mutually dispersed in spatial Query-Gram space—and the two could in principle be composed (InitNO for per-seed validity, SPECATTN-DIV for cross-seed diversity).

Table 16: **Multi-step sweep — moderate regime** (drift= 3.0, noise= 8.0). Shaded rows are SPECATTN-DIV.

Update geometry	Metric	$K=1$	$K=2$	$K=3$	$K=5$	$K=10$
Spherical Tangent (Ours)	Norm Drift ↓	0.000	0.000	0.000	0.000	0.000
	Vendi ↑	2.720	2.884	2.878	2.977	3.110
	Aesthetic ↑	5.267	5.188	5.172	5.102	5.021
Euclidean Additive (DivIn-style)	Norm Drift ↓	0.350	0.687	0.962	1.515	2.886
	Vendi ↑	2.727	2.891	2.942	3.066	3.165
	Aesthetic ↑	5.239	5.167	5.120	5.063	4.991
Euclidean + Post-hoc Rescale	Norm Drift ↓	0.000	0.000	0.000	0.000	0.000
	Vendi ↑	2.726	2.876	2.890	2.980	3.097
	Aesthetic ↑	5.258	5.180	5.157	5.108	5.034

Table 17: **Multi-step sweep — aggressive regime** (drift= 12.0, noise= 48.0). Shaded rows are SPECATTN-DIV. Latent Std given in parentheses for the $K=10$ endpoint.

Update geometry	Metric	$K=1$	$K=2$	$K=3$	$K=5$	$K=10$
Spherical Tangent (Ours)	Norm Drift ↓	0.000	0.000	0.000	0.000	0.000
	Vendi ↑	3.042	3.160	3.164	3.224	3.153
	Aesthetic ↑	5.093	4.986	4.898	4.814	4.744
Euclidean Additive (DivIn-style)	Norm Drift ↓	9.524	17.720	23.744	33.336	55.826
	Vendi ↑	3.233	3.191	3.106	2.896	2.296
	Aesthetic ↑	4.814	4.204	3.858	3.713	3.456
Euclidean + Post-hoc Rescale	Norm Drift ↓	0.000	0.000	0.000	0.000	0.000
	Vendi ↑	3.035	3.155	3.144	3.184	3.135
	Aesthetic ↑	5.099	4.982	4.905	4.794	4.693

DiverseFlow (Morshed & Boddeti, 2025) (2025) maximizes a Determinantal Point Process (DPP) log-volume kernel over DINO-ViT-B/8 features to encourage feature-space coverage. Crucially, it ascends this kernel at *every* reverse sampling step $t \in [T-1, 0]$, computing pairwise DINO embeddings and back-propagating through the denoiser at each step; this is a trajectory-stage intervention with substantial per-step latency. SPECATTN-DIV, by contrast, acts *once* strictly at $t=T$ with $\approx 3\%$ overhead and zero reverse-sampling cost (§4.7). The two occupy different points on the cost-diversity spectrum, and initialization-stage dispersal composes orthogonally with trajectory-stage guidance, as our CADs/IG/PG results demonstrate (Tables 1, 2).

TPSO (Anonymous, 2025) (2025) optimizes the text-prompt space for diversity. At the time of writing it has no publicly available code or released implementation (“code will be released upon acceptance”), and the paper does not specify all hyperparameters required for a faithful re-implementation; we therefore report it as a related method but do not attempt a potentially misleading re-implementation. We will incorporate a direct quantitative comparison once an official implementation becomes available.

Summary. All three methods target a different lever: InitNO targets per-image concept presence, DiverseFlow targets reverse-trajectory feature coverage, and TPSO targets prompt-space perturbation. SPECATTN-DIV uniquely targets *initialization-time inter-sample spatial-layout dispersal* on the Gaussian hypersphere, which is why it stacks orthogonally with trajectory guidance while adding negligible cost.

F HYPERPARAMETER SPECIFICATIONS

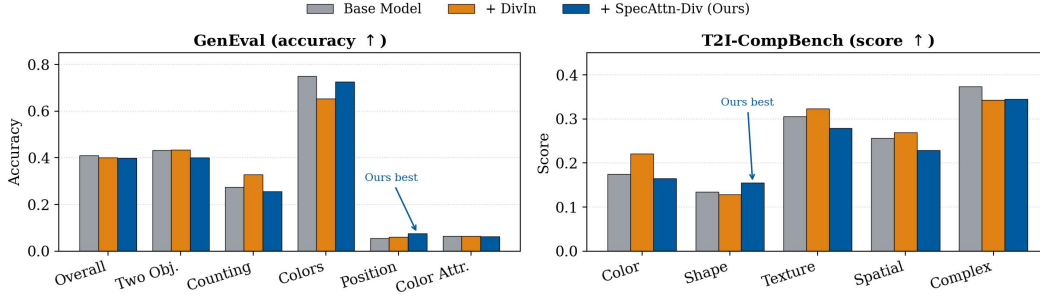


Figure 8: Compositional benchmark accuracy on GenEval (left) and T2I-CompBench (right), SD v1.4 (numbers in Table 6). SPECATTN-DIV preserves overall compositional integrity while achieving the best spatial **Position** accuracy (GenEval) and **Shape** score (T2I-CompBench), confirming that inter-sample layout dispersal does not degrade attribute binding or spatial relations.

Table 18: Default hyperparameters for SPECATTN-DIV across all experimental settings.

Parameter	Default value / setting
Curvature weight α	1.0
Repulsion weight β	2.0
Fidelity anchor weight γ	1.0
Hinge threshold τ_{fid}	-0.22
Finite-difference step ϵ	3.0
Tangent drift magnitude	3.0
Tangent noise magnitude	8.0
Langevin steps K	1
Precision mode	bf16 autocast (init probe) / FP16 (denoising)
Random seed	42 (deterministic PyTorch & cuDNN)